Hazeshaman

2026-10-13 · 23:00 MYT

MALAYSIA · HAZE RIGHT NOW

UNHEALTHY

Highest API

175

Pasir Gudang

Stations > 100

27 / 68

25 above 150

Stations at closure risk

13 / 68

highest: Pasir Gudang, Johor (83%)

Schools likely affected

2,616

closure chance × schools, state by state · 1,213,805 students

Model estimate from hourly DOE API – confirm with official APIMS and KPM announcements.

Tomorrow morning outlook

The dashboard in one screen



1. **Live badge** – green = the newest hourly DOE report is under 2 h old; amber/red = the feed is lagging or stale.
2. **Haze right now** – the sky, skyline and band name show how bad the air is across Malaysia at this hour.
3. **State tiles** – highest API per state, coloured by band. Click one to open that state.
4. **Live data / What-if** – Live follows the newest report; What-if lets you type your own readings.
5. **State filter** – the whole page follows it: "Malaysia" (all states) or one state. On a phone, use the round button at the bottom-right.
6. **National outlook** – tomorrow morning: stations at closure risk and the schools, students and teachers behind them.

STEP 2 · FILTER BY STATE

Malaysia or one state – on desktop and on your phone



Choose a state: the sky, the outlook, the schools and the station list all follow your choice. The station picker then unlocks:



On a phone the same list opens from the round button (bottom-right) as a bottom sheet:



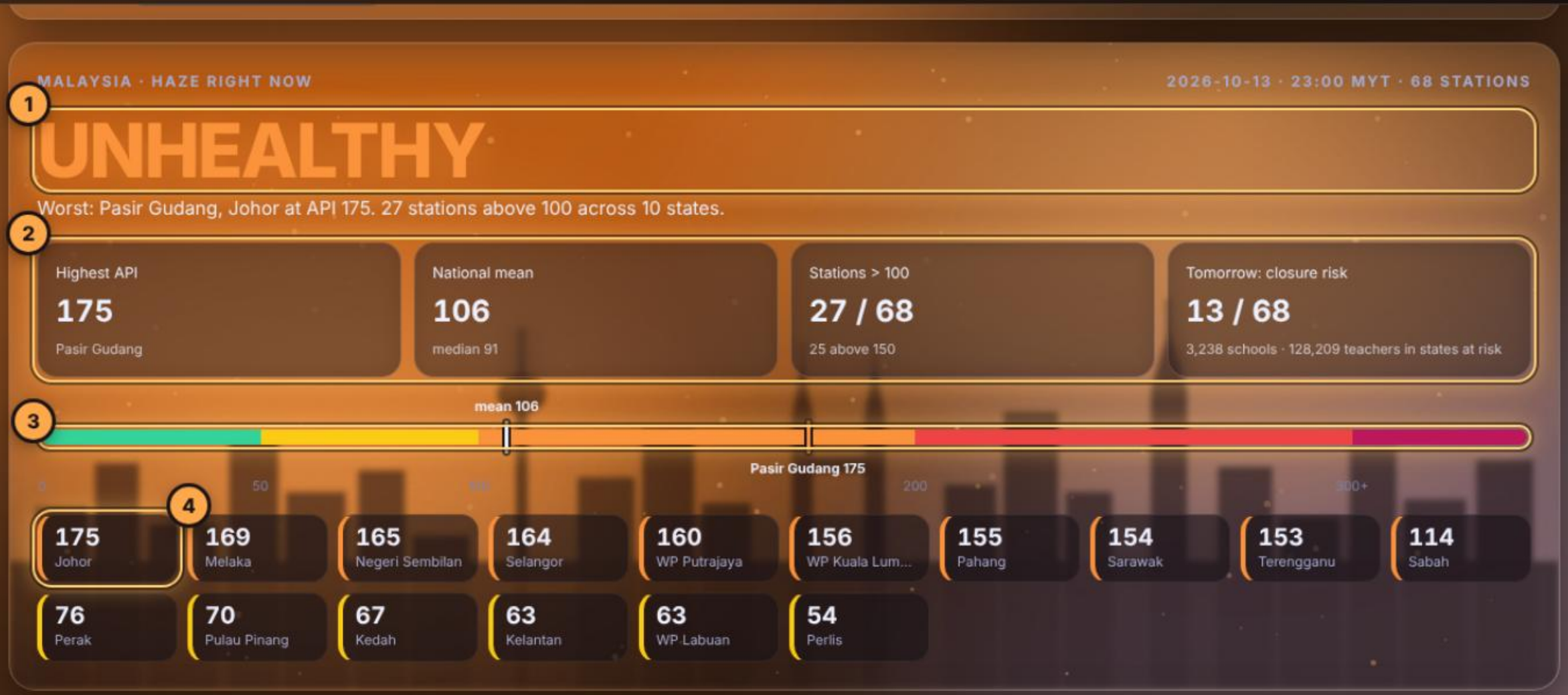
Round button stays in reach while you scroll



Tap a state; the page filters at once

1. **State button** – shows what the page is filtered to. Click it to open the list.
2. **State list** – "Malaysia" first, then every state. Each row shows its highest live API (coloured by band) and Open / Watch / Close status.
3. **Station picker** – after choosing a state, pick one of its stations or leave it on "Highest risk (auto)".
4. **Round button** (phones) – always bottom-right; opens the list.
5. **Bottom sheet** – tap a state. The link is shareable: add `#Johor` to the address.

Haze right now



1. **Band** – Malaysian API bands: Good 0–50 · Moderate 51–100 · Unhealthy 101–200 · Very unhealthy 201–300 · Hazardous 300+. The colour follows the *worst* station.
2. **Four numbers** – highest API, national mean, stations above 100 / 150, and tomorrow's closure-risk count.
3. **Band scale** – the white marker is the national mean; the coloured marker is the worst station.
4. **State tile** – the state's highest reading. The more haze, the thicker the particles and the blurrier the skyline.

National outlook and schools impacted

MALAYSIA · OUTLOOK FOR TOMORROW MORNING

68/68 stations reporting · latest hour 23:00 MYT

1 Stations at closure risk
13 / 68
highest: Pasir Gudang, Johor (83%)

Schools in states at risk
3,238
5 of 16 states · 1,437,339 students · 128,209 teachers

2 Schools likely affected
2,616
closure chance × schools, state by state · 1,213,805 students

schools monitored
10,238
4,818,596 students · 417,462 teachers · 20 public universities

State	Status (3)	Top station	Closure	7AM>150	Stns	# schools in area (5)
Johor	Close / PdPR	Pasir Gudang	83%	90%	7 / 8	1,195
Negeri Sembilan (4)	Close / PdPR	Nilai	79%	93%	2 / 3	479
Melaka	Close / PdPR	Bukit Rambai	78%	85%	1 / 3	318
Terengganu	Close / PdPR	Kemaman	73%	95%	1 / 4	506
Pahang	Close / PdPR	Rompin	48%	68%	2 / 5	740
WP Putrajaya	Watch	Putrajaya	12%	91%	0 / 1	27
Sarawak	Watch	Sri Aman	12%	77%	0 / 13	1,460
Selangor	Watch	Banting	9.8%	97%	0 / 6	945
WP Kuala Lumpur	Watch	Batu Muda	4.5%	92%	0 / 2	290
Sabah	Open	Tawau	0.1%	2.5%	0 / 6	1,298
Perak	Open	Seri Manjung	<0.1%	0.3%	0 / 5	1,105
Pulau Pinang	Open	Minden	<0.1%	0.2%	0 / 4	396
Kelantan	Open	Kota Bharu	<0.1%	0.1%	0 / 2	595
WP Labuan	Open	Labuan	<0.1%	0.1%	0 / 1	28
Kedah	Open	Sungai Petani	<0.1%	<0.1%	0 / 4	752
Perlis	Open	Kangar	<0.1%	<0.1%	0 / 1	104

Click a state to open its stations. A state's closure chance is its highest-risk station; 'Stns' = stations at risk / reporting; '# schools in area' = government primary + secondary schools in the state; 'Schools likely affected' = closure chance × those schools, summed over states.

1. **Stations at closure risk** – stations whose closure probability is above the cost-optimal trigger.
2. **Schools likely affected** – closure chance × the number of schools in each state, added up (students and teachers shown underneath).
3. **Status** – Close / PdPR, Watch or Open per state, worst first. A state's chance is its highest-risk station.
4. **Click a row** to filter the page to that state and see its stations.
5. **# schools in area** – government primary + secondary schools in the state (state totals, MOE via OpenDOSM).

1



LARKIN · JOHOR · RECOMMENDATION

Close tomorrow · switch to PdPR

Closure risk 71% is above the cost-optimal trigger of 25%. Notify parents and hostels; re-check at 8 PM and 6 AM. ⚠ Inputs are outside the range seen in training – indicative only.

2

7 AM API > 100	7 AM API > 150	API > 180 in school hours	Streak rule
98%	**83%**	**7.4%**	**69%**
unhealthy	very unhealthy	closure trigger	tomorrow mean > 150

3

SCHOOLS IMPACTED · JOHOR

Schools in Johor	Students and teachers	If schools close	Likely affected tomorrow
1,195	**581,184**	**1,195**	**989**
govt primary + secondary · +2 public universities	48,876 teachers · 11.9 pupils per teacher	schools · 581,184 students · 48,876 teachers affected	83% closure chance × 1,195 schools

Closure chance is the state's highest-risk station (Pasir Gudang). "Likely affected tomorrow" = that chance × the number of schools in Johor; "If schools close" is the full count affected when a state-wide closure is ordered. School counts are state-level totals from OpenDOSM/data.gov.my: schools_district, enrolment_school_district, teachers_district (2025; government primary + secondary schools; IPTA public universities listed separately).

4

THREE SCENARIOS FOR TOMORROW MORNING

Best · improving	Base · unstable	Worst · closure
0.5%	**29%**	**71%**
7 AM reading ≤ 100. Announce open early; families can plan.	Unhealthy but below closure trigger. Open with restrictions; keep PdPR ready.	API > 180 in school hours, or 3rd straight day > 150. Close / PdPR.

1 **Recommendation** – open / open with restrictions / close-PdPR, with the closure ring and why.

2 **Probabilities** – 7 AM API above 100 and 150, API above 180 in school hours, and the 3-day streak rule (KPM's two closure triggers).

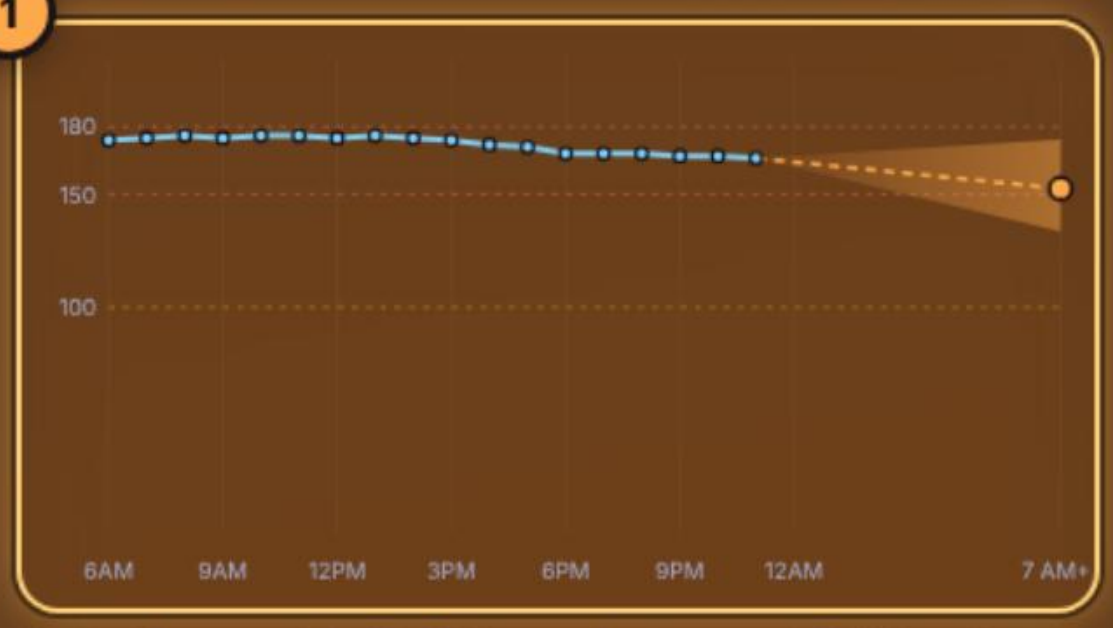
3 **Schools impacted (state level)** – schools, students and teachers in the state. "If schools close" = everyone affected by a state-wide closure; "Likely affected tomorrow" = closure chance × schools.

4 **Three scenarios** – best (improving), base (unstable), worst (closure). The framed one is the most likely.

STEP 6 · WHEN TO ANNOUNCE

The best announcement time reacts to your preferences

1



Dashed fan = model's 10–90% range for tomorrow's 7 AM reading (median 153).

2



★ 3 PM minimises total expected cost at your preferences (missed closure = 8× needless closure; each hour later = 5% of baseline). Click a bar to explore that hour in What-if.

Now drag "Cost of each hour later" from 5% to 0% (delay is free) – the ★ moves from the earliest hour to the latest:

DECISION PREFERENCES

Wrongly staying open costs × a needless closure **8**

3

ost of each hour later of baseline **0%**

4



★ 9 PM minimises total expected cost at your preferences (missed closure = 8× needless closure; each hour later = 0% of baseline). Click a bar to explore that hour in What-if.

1 **Trend + fan** – today's hourly API and the model's 10–90% range for tomorrow 7 AM.

2 **Cost bars** – wrong-call cost (missed closure vs needless closure) plus lateness cost for each candidate hour; ★ = cheapest. Click a bar to explore that hour.

3 **Decision preferences** (left panel) – how bad a missed closure is, and what each hour of lateness costs.

4 **Result** – with free delay the ★ moves to 9 PM because later hours forecast better; with costly delay the earliest hour wins.

STEP 7 · TRY YOUR OWN NUMBERS

What-if mode

MODE (1)

Live data | What-if

Model hour (2) — 11 PM

12 PM · 1 PM · 2 PM · 3 PM · 4 PM · 5 PM · 6 PM · 7 PM · 8 PM · 9 PM · 10 PM · 11 PM

What-if: edit any reading below

API at 8 AM (3) — 105

API at 12 PM — 120

API at 8 PM — 135

API at 11 PM — 150

Yesterday's daily mean — 163

Prior consecutive days with mean > 150 (4)

0 · 1 · 2 · 3+

Editing Larkin, Johor.

DECISION PREFERENCES

Wrongly staying open costs × a needless closure — 8

Cost of each hour later of baseline — 5%

(5) **Result**

5%
closure risk

LARKIN · JOHOR · RECOMMENDATION

Open with restrictions · PdPR on standby

Closure risk 4.6% is below the trigger (25%) but not negligible. Cancel outdoor activity and confirm at 8 PM / 6 AM. ⚠ Inputs are outside the range seen in training – indicative only.

7 AM API > 100	7 AM API > 150	API > 180 in school hours	Streak rule
100%	90%	4.6%	<0.1%
unhealthy	very unhealthy	closure trigger	needs ≥2 prior days > 150

SCHOOLS IMPACTED · JOHOR

Schools in Johor	Students and teachers	If schools close	Likely affected tomorrow
1,195	581,184	1,195	55
govt primary + secondary · +2 public universities	48,876 teachers · 11.9 pupils per teacher	schools · 581,184 students · 48,876 teachers affected	4.6% closure chance × 1,195 schools

Closure chance is the state's highest-risk station (Larkin, editable here). "Likely affected tomorrow" = that chance × the number of schools in Johor; "If schools close" is the full count affected when a state-wide closure is ordered. School counts are state-level totals from OpenDOSM/data.gov.my: schools_district, enrolment_school_district, teachers_district (2025; government primary + secondary schools; IPTA public universities listed separately).

THREE SCENARIOS FOR TOMORROW MORNING

Best · improving	Base · unstable	Worst · closure
0.4%	95%	4.6%
7 AM reading ≤ 100. Announce open early; families can plan.	Unhealthy but below closure trigger. Open with restrictions; keep PdPR ready.	API > 180 in school hours, or 3rd straight day > 150. Close / PdPR.

Best 0.4% · Base 95% · Worst 4.6%

TODAY'S TREND → TOMORROW 7 AM

180

BEST ANNOUNCEMENT TIME

wrong-call cost · lateness cost

47%

35%

1 **What-if** – switch here; sliders unlock. Switch back to Live data at any time.

2 **Model hour** – pick the hour you would decide at (outlined = candidate announcement times).

3 **Readings** – set API at 8 AM, noon, three hours earlier and now, plus yesterday's mean.

4 **Streak** – how many prior days had a mean above 150 (KPM's three-day rule).

5 **Result** – the recommendation and every probability recompute instantly.

Same dashboard, stacked for small screens



Haze right now



National outlook



Station recommendation